11AG-MATH-FINAL — MARKING SCHEME

Teacher copy — not for distribution. Grade 11AG · 26/10/2026 · Total 100 marks

Section A — Easy (6 questions × 4 marks = 24 marks)

Answer all six questions.

1. Find $\lim_{x \to 2} (3x^2 - 5)$. (4 marks)

Answer: 7

2. Find $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$. (4 marks)

Answer: 6

3. Find $\lim_{x \to \infty} \dfrac{5x^2 + 1}{2x^2 - x}$. (4 marks)

Answer: $\dfrac{5}{2}$

4. Differentiate $y = 4x^5$. (4 marks)

Answer: $20x^4$

5. Differentiate $y = \sin x + e^x$. (4 marks)

Answer: $\cos x + e^x$

6. Differentiate $y = \ln x - \cos x$. (4 marks)

Answer: $\dfrac{1}{x} + \sin x$

Section B — Medium (6 questions × 6 marks = 36 marks)

Answer all six questions. Show your working.

7. Find $\lim_{x \to 1} \dfrac{x^2 + 2x - 3}{x - 1}$. (6 marks)

Answer: 4

8. Find $\lim_{x \to \infty} \dfrac{3x^3 - 2x}{6x^3 + 5}$. (6 marks)

Answer: $\dfrac{1}{2}$

9. Differentiate $y = (2x + 1)^5$. (6 marks)

Answer: $10(2x + 1)^4$

10. Differentiate $y = x^2 e^x$, giving your answer in factorised form. (6 marks)

Answer: $xe^x(x + 2)$

11. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. (6 marks)

Answer: $4\sec^2 4x$; gradient 4

12. Differentiate $y = \ln\left(3x^2 + 1\right)$. (6 marks)

Answer: $\dfrac{6x}{3x^2 + 1}$

Section C — Hard (3 questions, 40 marks)
Answer all three questions. Show every step of your working.

13. Evaluate the following limits: (13 marks)

(a) $\displaystyle\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$ (6 marks)

Answer: $\dfrac{1}{4}$

(b) $\displaystyle\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - x\right)$ (7 marks)

Answer: $\dfrac{3}{2}$

14. Differentiate, simplifying your answer where possible: (13 marks)

(a) $y = \sqrt{x^2 + 9}$ (6 marks)

Answer: $\dfrac{x}{\sqrt{x^2 + 9}}$

(b) $y = \ln(\cos x)$ (7 marks)

Answer: $-\tan x$

15. A curve has equation $y = x^2 \sin x$. (14 marks)

(a) Find $\dfrac{dy}{dx}$. (7 marks)

Answer: $2x \sin x + x^2 \cos x$

(b) Find $\dfrac{d^2y}{dx^2}$. (7 marks)

Answer: $2 \sin x + 4x \cos x - x^2 \sin x$

— End of the exam —